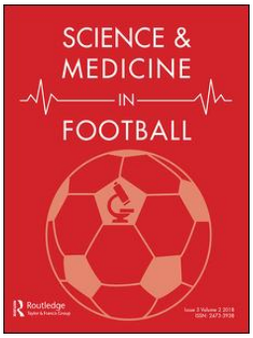


Routledge
Taylor & Francis Group

Science and Medicine in Football

ISSN: (Print) (Online) Journal homepage: https://www.tandfonline.com/loi/rsmf20

Is there a relationship between lower-extremity injuries and foot postures in professional football players? A prospective cohort study

Afsaneh Safar Cherati , Salman Khalifeh Soltani , Navid Moghadam , Bahar Hassanmirzaei , Zohreh Haratian , Shayesteh Khalifeh Soltani & Meisam Rezaei

To cite this article: Afsaneh Safar Cherati , Salman Khalifeh Soltani , Navid Moghadam , Bahar Hassanmirzaei , Zohreh Haratian , Shayesteh Khalifeh Soltani & Meisam Rezaei (2020): Is there a relationship between lower-extremity injuries and foot postures in professional football players? A prospective cohort study, Science and Medicine in Football, DOI: 10.1080/24733938.2020.1870711

To link to this article: https://doi.org/10.1080/24733938.2020.1870711

Accepted author version posted online: 29 Dec 2020.

Submit your article to this journal

Article views: 13

View related articles

View Crossmark data

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=rsmf20

Check for updates

Title page

Is there a relationship between lower-extremity injuries and foot postures in professional football players? A prospective cohort study

Afsaneh Safar Cherati, MD.[1] Salman Khalifeh Soltani, MD.[1,2*], Navid Moghadam, MD.[3], Bahar Hassanmirzaei, MD.[3,2], Zohreh Haratian, MD.[2], Shayesteh Khalifeh Soltani MD.[4], Meisam Rezaei MD.[1]

1. Iran University of Medical Sciences, Department of Sport and Exercise Medicine, Tehran, Iran

2. Iran Football Medical Assessment and Rehabilitation Center – IFMARC, Tehran, Iran

3. Sports Medicine Research Center, Neuroscience Institute, Tehran University of Medical Sciences, Tehran, Iran

4. Neuromusculoskeletal Research Center, Iran University of Medical Sciences, Tehran, Iran.

* Corresponding Author:

Salman Khalifeh Soltani

Iran University of Medical Sciences, Department of Sport and Exercise Medicine, Tehran, Iran

Iran Football Medical Assessment and Rehabilitation Center – IFMARC, Tehran, Iran

Address: Sports Medicine Department, Rasool Hospital, Niayesh St. Satarkhan St. Tehran, Iran.

Postal Code: 1445613131

Tel: +989133091860

Email: salmankhs@outlook.com

ACCEPTED MANUSCRIPT

Is there a relationship between lower-extremity injuries and foot postures in professional football players? A prospective cohort study

Abstract

Lower extremity injuries are an ongoing concern for professional football players. This study aimes to evaluate the relationship between foot posture and lower extremity injuries in professional football players. In this prospective cohort study, 420 male players of the Iran Premium football league were evaluated during the 2015-2016 season. The players were assessed for the subtalar joint position and their foot types, based on optical and static foot scans, and also their foot posture index (FPI). The trained club physicians recorded all injuries during the season. The analyzed data of 244 players showed the highest rate of lower extremity injury in hamstrings, ankle, and groin, respectively. These injuries led to 46% of time loss. The probability (relative risk) of hamstrings injuries was higher among pronated players based on all four evaluation methods (static, optic, FPI, and visual examination, 2.1, 1.8, 1.8, and 2.3, respectively). Medial Collateral Ligament (MCL) injuries were associated with subtalar joint abnormality defined by visual observation. An increased relative risk of leading-to-absence injuries among the flat foot group was significant based on optic scanning, FPI, and visual observation. Conclusion: Abnormal foot postures in professional football players may increase the risk of hamstring and MCL injuries as well as time loss due to lower exteremity injuries.

Key Words: Foot posture, Lower extremities injuries, football, FPI, Foot scan

ACCEPTED MANUSCRIPT

Introduction

According to the Fédération Internationale de Football Association (FIFA) study, the total number of football players assumed 265 million worldwide [1], among which around 200,000 are professional or semiprofessional [2]. An average number of injuries for a player is two per season [3]. The most common injuries among football players are those related to the lower extremities. Based on FIFA evaluation, about $30 billion is spent on injury management of football players annually [4].

Studies have shown that biomechanical abnormalities are significantly associated with lower extremity injuries. This is illustrated by the probable relationship between increased subtalar joint inversion and Achilles tendinopathy [5] or pronated foot type with medial tibial stress syndrome [6-8]. Identification of risk factors provides appropriate strategies for the development of a prevention program. The literature has already stated that reduced flexibility of the hamstring or quadriceps muscles and previous injuries like a hamstring injury, knee joint injury, and groin injury are main intrinsic risk factors of the lower extremity lesions in football/soccer players [9-12].

Foot posture playes a critical role in normal lower limb kinetics and kinematics [13-15], and foot pathologies can lead to a significantly higher risk of lower limb injury [16-18]. However, there is a lack of high-quality trials. Consequently, this study aimed to determine the prevalence of lower extremity injuries prospectively and evaluate the correlation with the foot's pathomechanics in male professional football players in the Iran Premium Football League during the 2015-2016 season.

ACCEPTED MANUSCRIPT

Materials and Methods

Study sample and study period

This prospective cohort study evaluates the male professional football players during the 2015-2016 season (September to May). The assessment was carried out by the biomechanical department of Iran Football Medical Assessment and Rehabilitation Center (IFMARC). All the 16 clubs of Iran Football Pro League participated in the Pre-Participation Medical Assessment (PCMA) study. The injuries of lower extremities during the season were recorded and analyzed. The age range of the players included is between 15 and 35 years.

Inclusion criteria:

All the adult players who play in Iran's men's football premium league during the 2015-2016 season

Exclusion criteria:

Leaving the premium football league of Iran during the 2015-2016 season

Failure to report the injuries by the club physicians.

The ethics committee of the Iran University of Medical Sciences and the Medical Committee of Iran Football Federation approved the research. Signed written informed consents were obtained from all included players (420 players).

Baseline Examinations and assessment

There are different clinical methods for determining the foot posture, out of which the combination of the four most common and valid ones was chosen in the study to improve the

ACCEPTED MANUSCRIPT

validity of the results. Their test quality properties are proven with good to excellent inter-rater/intra-rater/test-retest reliability and validity [19-30].

1. The most practical method for foot posture assessment is visual observation. All observations were made by looking at the heels from anterior and posterior aspects and at the medial side's plantar arch. Each foot was evaluated based on the criteria mentioned in Laurie K. et al. 's study and assigned as pronated, supinated, or normal foot [19]. The examiner (certified Sports Medicine specialists) was not aware of the foot scan results.

2. Foot posture index (FPI), a clinical diagnostic tool with the quantifiable result, identifies six measurements. Each component is scored from -2 to +2, to sum up, -12 to +12, indicating extreme supination to the extreme of pronation, respectively [31]. FPI is converted to categorical variables in a scale spectrum. Supinated foot and the pronated foot is recognized with <-1 and >+6 criteria, respectively. Within the -1 to +6 range the foot is perceived as normal foot.

3. Foot types according to the optical foot scan by a Kortech device. Based on the report of the assessment, the length of the footprint excluding the toes (L), is divided into equal thirds A, B, and C: A forefoot, B midfoot, and C heel. Arch Index (AI) was calculated as the area of the middle third of the footprint divided by the entire footprint area (AI = B/A + B + C) [32]. Three categories of normal (0.21 to 0.28), high (<0.21), and low (>0.28) AI Score were defined.

4. A static pressure foot scan was assessed by Ghiwell South Korea Force Plate (36.9 cm x 43.6 cm with 2,288 sensors of 1.4 /cm2 resolution to record 40 frames per second). This standard foot scan has been measured in the pre-competition medical assessment (PCMA) of international elite football players before the 2006 FIFA World Cup [33]. Certified Sports Medicine specialists did all the examinations. In the aim of determining the foot posture, participants were asked to stand

ACCEPTED MANUSCRIPT

on their usual bipedal stance, and the pressure under each of their feet was recorded on the Matscan system. Modified Arch Index (MAI) value was calculated through the formula:

$MAI = Pressure_B/(Pressure_A + Pressure_B + Pressure_C)$ [34].

Injury record

The Club physicians were requested to record and report any injuries, leading to at least one-day absence from match/training sessions. The other criteria were gathered, categorized, and standardized by the researchers of this study to prevent any bias resulting from different points of views. Demographic data included age, previous injuries, treatment history, background problems were obtained based on PCMA form page 2-4 [33]. The provided information includes the date, type, and location of the injury. The severity of the injuries was categorized based on the duration of absence from training mild to moderate (<28 days) and severe (≥ 28 days).

Lower extremitiy injures including the groin, quadriceps muscle, hamstring muscle, knee (Anterior Cruciate Ligament (ACL), Posterior Cruciate Ligament (PCL), Medial Collateral Ligament or Medial Collateral Ligament (MCL), Lateral Collateral Ligament (LCL), medial meniscus, lateral meniscus, non-ligamentous, non-meniscus), and ankle (ligamentous, non-ligamentous) were loged. In this study, injuries were recorded according to the soccer injury surveillance consensus [35].

Dispite simultaneous injuries, re-injuries were not included in the total injury count in order to specify better the relation of abnormal foot posture and injury occurrence, considering the hypothesis that injured players may be more susceptible to other types of injruies.

Statistical Analyses

ACCEPTED MANUSCRIPT

Data were analyzed by SPSS v. 22.0. For the association of foot type (based on different methods) and injuries, we used relative risk with 95% CI. As we considered 95% CI, the significance level was 0.05. Our analysis is player-based for both foot posture and injuries evaluation. The relative risks of the injuries associated with each abnormal foot posture (pronated, supinated and both together as abnormal) based on each assessment method (Optic, Static, Visual, and FPI) were calculated.

We also used a nonparametric bivariate Spearman test to assess the correlation between foot posture types based on four methods (Optic, Static, Visual, and FPI) with each other.

Results

The information of 244 players out of 420 players was recorded by club physicians; 176 players have left the study either as a result of transferring to other leagues or difficulties in injury records (Figure 1).

Foot Posture

The foot examination of 244 football players to identify their foot posture type based on four different methods of investigation is shown in Figure 2.

Injury record

Total injuries:

One hundred fifty-five injuries were recorded, including at least one-day absence from training or match, of which most common injuries were hamstring muscle injuries (n=41, 26%), ankle ligament injuries (n=34, 22%) and adductor injuries (n=25, 16%) respectively. Table 1 and

ACCEPTED MANUSCRIPT

figure 3 presents the prevalence of lower extremity injuries recorded based on PCMA form among all injuries.

Total players injured:

The total number of players who sustained lower extremity injuries were 111(45.49% of total players), of which 78 players (31.97% of total players) experienced at least one mild to moderate lower extremity injury (with <28 absence days) and 33 players (13.52 % of total players) suffered at least one severe lower extremity injuries (with ≥ 28 absence days). we had 44 players recorded with simultaneous more than one injury.

Table 1 demonstrate the prevalence of lower extremity injuries recorded based on PCMA form among all players.

Foot Posture and injuries correlation

Table 2 illustrates the relative risk with a 95% confidence interval (CI) of injuries among abnormal (both pronated and supinated as abnormal) foot posture types of professional football players based on four different examination methods.

Based on the optical and static pressure foot scan examinations, there is a relationship between pronation and hamstrings injuries. The same results were identified based on FPI evaluation. Football players` FPI has a normal distribution (mean = 3.0 and SD=5.0) based on Kolmogorov-Smirnov and Shapiro-Wilk tests with the p-values of 0.149 and 0.952, respectively.

The supination, based on visual observation, had no relationships with any injuries. Still, players with abnormal foot postures (supination or pronation) have a higher relative risk of injury in the hamstring or MCL.

ACCEPTED MANUSCRIPT

Professional football players with pronated feet based on visual observations were at a higher risk of hamstrings injuries (Figure 4). Pronation also is associated with milder to moderate injuries (fewer than 28 days of absence) compared to severe injuries (more than 28 days of absence).

Briefly, based on our results, pronated foot posture based on all four assessment methods can be a predictor of hamstring injuries and the absence of fewer than 28 days according to the three methods as well. Abnormal foot posture based on the visual examination (one method) also can be a predictor of MCL and hamstring injuries.

Relative risk between each foot posture type(pronated/supinated) based on all four examinations and lower limb injuries are demonstrated in figures 4 to 7.

Comparison of foot posture assessment methods

The four methods used in the study were compared with each other. The Spearman test showed a significant *strong* positive paired correlation ($p<0.001$- $r>0.9$) for FPI-static, optic-static, and FPI-optic tests, as well as significant *moderate* correlation ($p<0.001$- $r>0.3$) for visual examination compared with the optic, static and FPI tests.

Discussion

This prospective study aimed to evaluate the association of different foot types and lower extremity injuries in Iranian professional male football players. It is known that, different foot postures are accompanied by an alteration in biomechanics, kinetics, and foot function [36], even mediated by foot orthosis [37]. This study indicates that such biomechanical considerations can

ACCEPTED MANUSCRIPT

be applied to football players, too. We were able to show that abnormal foot postures in professional football players may increase the risk of hamstring and MCL injuries as well as time loss due to lower exteremity injuries.

The average FPI score for the normal healthy population is 2.4 (slightly pronated)[38]. However, the mean FPI in our cohort of professional football players was 3.0±5.0 regardless of injuries. This number is almost the same in other measurements (2.1, 2.0, and 2.2, based on the optical scan, static pressure scan, and visual observation methods, respectively). We found the distribution of foot postures in the Iran Premium league as 32.0%, 17.2%, and 50.8% for pronation, supination, and normal foot posture. The mean FPI has already been calculated as 2.9 ± 2.8, 3.9 ± 4.1, and -0.4 ± 6.9 in runners, basketball, and handball players, respectively, with significant differences among these groups (P = .008)[39]. This is confirmed by the results of a previous study in indoor football (Futsal) players of the Iranian premier league (4.9±2.8 and 4.7±3.0 for right and left foot)[40]. The outdoor sports players' foot postures, such as runners or grass football players, are more similar to the healthy population compered to the indoor players.

Our results suggest that having extremes of foot posture predisposes athletes to a higher risk of mild to moderate lower extremity injuries.

The comparison of foot posture by four different methods is one novelty of our study. The distribution of foot postures based on visual observation, foot posture indices (FPIs), optical foot scan, and static pressure foot scan is correlated significantly and positively. We observed the strongest correlations for FPI-Static, FPI-Optic, and Optic-Static comparisons. Visual examination showed a lower positive correlation with the other three tests. Although it is a reliable means to classify the foot type[19], the lower correlation with FPI, Optic, and Static evaluations may refer to the observer's visual measurement errors with less quantified values.

ACCEPTED MANUSCRIPT

Other studies also demonstrated this association between clinical and radiological methods ($p<.05$)[32]. Pronated foot based on Optic, FPI and visual evaluations were more associated with injury report.

Among non-athletes, there are significant relationships between foot posture abnormality and over-use injuries such as pronated foot postures and lateral ankle sprain[41], patellofemoral pain syndrome[42], metatarsal, and lower limb stress fractures[5, 43]. This study also showed a different aspect for the professional football players as the probability of hamstring injuries could be higher among pronated players, and the abnormality may increase the incidence of MCL injuries, too.

The fundamental cause may be finding hamstring tightness in players with pronated foot posture and subsequent proneness to hamstring muscle injury in them. Regarding the available evidence, a pronated foot kinetically alters the lower limbs, lumbopelvic, and trunk muscles activation during the gait, possibly to compensate for the overload of the medial longitudinal arc[44-48]. Anti-pronation footwear is reported to be effective in increasing peak knee extensor, hip abductor and hip internal rotator moments during running; similarly, reduced hip adduction and knee internal rotation angles during a step-up task as well as during walking[49-53]. Lesser knee extensor and hip abductor moments may explain the hamstring overload and tightness. Likewise, the MCL injuries can be provoked by a higher knee internal rotation angle and transverse rotational motion, which leads to a higher medial load.

Further studies should focus on evaluating of hamstring and knee injuries in players with pronated foot posture and assessment of corrective interventions on preventing injuries in this group of players. As in some situations, foot posture is modifiable by using external braces or even undertaking surgery; these outcomes may have implications for preventing injuries in elites.

ACCEPTED MANUSCRIPT

There is also a correlation between supinated feet and injuries such as tibial and femoral stress fractures[43] among non-sportspersons, but it has not been demonstrated in professional football players so far and similarly not in our cohort.

Despite similar foot posture distribution, it is assumed that the functional pattern of foot postures is different in professional football players than in the non-sport population. However, regardless of the type and severity, the foot posture can predict injuries in both groups.

Although the literature is not conclusive, the results show the time-loss injuries among the players with a flat foot are more likely. Moreover, our study asserted that the type of sustained damage is probably related to foot classifications, which contrasts with Bassett's research [54]. Consequently, more prospective data are required to clarify the association between foot posture and injury risk in football players.

A further biomechanical study is suggested to explain how pronated foot posture in a football player can lead to hamstring and MCL injuries. It is also recommended to evaluate the injury reduction rate of corrective interventions for players with a pronated foot posture. Innovative prospective research is needed to identify other posture risk factors for developing elite injuries in other sports.

Limitations

This study's main limitation was the possible inter-rater and intra-rater variability in recording injuries and observational assessments.

ACCEPTED MANUSCRIPT

Limited number of players evaluated(244), single season and league also may influence the conclusion's generalizability.

Practical Implication

Foot posture assessment of professional football players, regardless of the evaluation method, make sense and correction of abnormal foot posture may have the potential to reduce the risks.

Acknowledgment

The authors would like to acknowledge the following: Iran Football League Organization and all 16 Team Physicians whom this work could not be done without their full contribution.

Declaration of interest statement

The authors declare no conflict of interest or involvement in any organization or entity with any financial interest in the subject matter or materials discussed in this manuscript.

References

1. Kunz, M., 265 million playing football. FIFA magazine, 2007: p. 10-15.
2. Arliani, G. G., Belangero, P. S., Runco, J. L., & Cohen, M., The Brazilian Football Association (CBF) model for epidemiological studies on professional soccer player injuries. Clinics (Sao Paulo), 2011. 66(10): p. 1707-1712.
3. Ekstrand J, Hägglund M, Waldén M., Injury incidence and injury patterns in professional football: the UEFA injury study. Br J Sports Med, 2010: p.60582.
4. Caine, D.J. and N. Maffulli, Epidemiology of pediatric sports injuries: Team sports. Vol. 2. 2005.

ACCEPTED MANUSCRIPT

5. Kaufman KR, Brodine SK, Shaffer RA, Johnson CW, Cullison TR., The effect of foot structure and range of motion on musculoskeletal overuse injuries. Am J Sports Med, 1999. 27(5): p. 585-593.
6. Bennett JE, Reinking MF, Pluemer B, Pentel A, Seaton M, Killian C., Factors contributing to the development of medial tibial stress syndrome in high school runners. J Orthop Sports Phys Ther, 2001. 31(9): p. 504-510.
7. Willems TM, De Clercq D, Delbaere K, Vanderstraeten G, De Cock A, Witvrouw E., A prospective study of gait related risk factors for exercise-related lower leg pain. Gait posture, 2006. 23(1): p. 91-98.
8. Yates, B. and S. White, The incidence and risk factors in the development of medial tibial stress syndrome among naval recruits. Am J Sports Med, 2004. 32(3): p. 772-780.
9. Arnason A, Sigurdsson SB, Gudmundsson A, Holme I, Engebretsen L, Bahr R., Risk factors for injuries in football. Am J Sports Med, 2004. 32(1 suppl): p. 5S-16S.
10. Bradley, P.S. and M.D. Portas, The relationship between preseason range of motion and muscle strain injury in elite soccer players. J Strength Cond Res, 2007. 21(4): p. 1155-1159.
11. Hägglund M, Waldén M, Ekstrand J., Previous injury as a risk factor for injury in elite football: a prospective study over two consecutive seasons. Br J Sports Med, 2006. 40(9): p. 767-772.
12. Witvrouw E, Danneels L, Asselman P, D'Have T, Cambier D. , Muscle flexibility as a risk factor for developing muscle injuries in male professional soccer players. Am J Sports Med, 2003. 31(1): p. 41-46.
13. Buldt AK, Allan JJ, Landorf KB, Menz HB., The relationship between foot posture and plantar pressure during walking in adults: a systematic review. Gait posture, 2018. 62: p. 56-67.
14. Maharaj JN, Cresswell AG, Lichtwark GA., Foot structure is significantly associated to subtalar joint kinetics and mechanical energetics. Gait posture, 2017. 58: p. 159-165.
15. Song JI, Hillstrom HJ, Secord D, Levitt J., Foot type biomechanics. comparison of planus and rectus foot types. J Am Podiatr Med Assoc, 1996. 86(1): p. 16-23.
16. Cain LE, Nicholson LL, Adams RD, Burns J. , Foot morphology and foot/ankle injury in indoor football. J Sci Med Sport, 2007. 10(5): p. 311-319.
17. Neal BS, Griffiths IB, Dowling GJ, Murley GS, Munteanu SE, Smith MM, Collins NJ, Barton CJ., Foot posture as a risk factor for lower limb overuse injury: a systematic review and meta-analysis. J Foot Ankle Res, 2014. 7(1): p. 55.
18. Tong, J.W. and P.W. Kong, Association between foot type and lower extremity injuries: systematic literature review with meta-analysis. J Orthop Sports Phys Ther, 2013. 43(10): p. 700-714.
19. Dahle LK, Mueller M, Delitto A, Diamond JE., Visual assessment of foot type and relationship of foot type to lower extremity injury. J Orthop Sports Phys Ther, 1991. 14(2): p. 70-74.
20. Menz HB, Fotoohabadi MR, Wee E, Spink MJ., Visual categorisation of the arch index: a simplified measure of foot posture in older people. J Foot Ankle Res, 2012. 5(1): p. 10.
21. Cowan DN, Robinson JR, Jones BH, Polly Jr DW, Berrey Jr BH., Consistency of visual assessments of arch height among clinicians. Foot Ankle Int, 1994. 15(4): p. 213-217.
22. Billis E, Katsakiori E, Kapodistrias C, Kapreli E., Assessment of foot posture: Correlation between different clinical techniques. The foot, 2007. 17(2): p. 65-72.

ACCEPTED MANUSCRIPT

23. Menz, H.B., Alternative techniques for the clinical assessment of foot pronation. J Am Podiatr Med Assoc, 1998. 88(3): p. 119-129.
24. Razeghi, M. and M.E. Batt, Foot type classification: a critical review of current methods. Gait posture, 2002. 15(3): p. 282-291.
25. Keenan AM, Redmond AC, Horton M, Conaghan PG, Tennant A., The Foot Posture Index: Rasch analysis of a novel, foot-specific outcome measure. Arch Phys Med Rehabil, 2007. 88(1): p. 88-93.
26. Morrison, S.C. and J. Ferrari, Inter-rater reliability of the Foot Posture Index (FPI-6) in the assessment of the paediatric foot. J Foot Ankle Res, 2009. 2(1): p. 26.
27. Cornwall MW, McPoil TG, Lebec M, Vicenzino B, Wilson J., Reliability of the modified foot posture index. J Am Podiatr Med Assoc, 2008. 98(1): p. 7-13.
28. Mentiplay BF, Clark RA, Mullins A, Bryant AL, Bartold S, Paterson K., Reliability and validity of the Microsoft Kinect for evaluating static foot posture. J Foot Ankle Res, 2013. 6(1): p. 14.
29. Commean PK, Mueller MJ, Smith KE, Hastings M, Klaesner J, Pilgram T, Robertson DD., Reliability and validity of combined imaging and pressures assessment methods for diabetic feet. Arch Phys Med Rehabil, 2002. 83(4): p. 497-505.
30. Zammit GV, Menz HB, Munteanu SE., Reliability of the TekScan MatScan® system for the measurement of plantar forces and pressures during barefoot level walking in healthy adults. J Foot Ankle Res, 2010. 3(1): p. 11.
31. Redmond AC, Crosbie J, Ouvrier RA., et al., Development and validation of a novel rating system for scoring standing foot posture: the Foot Posture Index. Clinical biomechanics, 2006. 21(1): p. 89-98.
32. Murley GS, Menz HB, Landorf KB., A protocol for classifying normal-and flat-arched foot posture for research studies using clinical and radiographic measurements. J Foot Ankle Res, 2009. 2(1): p. 22.
33. Dvorak J, Grimm K, Schmied C, Junge A., Development and implementation of a standardized precompetition medical assessment of international elite football players-2006 FIFA World Cup Germany. Clin J Sport Med, 2009. 19(4): p. 316-321.
34. Riskowski JL, Dufour AB, Hagedorn TJ, Hillstrom HJ, Casey VA, Hannan MT., Associations of foot posture and function to lower extremity pain: results from a population based foot study. Arthritis Care Res, 2013. 65(11): p. 1804-1812.
35. Fuller CW, Ekstrand J, Junge A, Andersen TE, Bahr R, Dvorak J, Hägglund M, McCrory P, Meeuwisse WH., Consensus statement on injury definitions and data collection procedures in studies of football (soccer) injuries. Scand J Med Sci Sports, 2006. 16(2): p. 83-92.
36. Farahpour N, Jafarnezhadgero A, Allard P, Majlesi M., Muscle activity and kinetics of lower limbs during walking in pronated feet individuals with and without low back pain. J Electromyogr Kinesiol, 2018. 39: p. 35-41.
37. Bonifácio D, Richards J, Selfe J, Curran S, Trede R., Influence and benefits of foot orthoses on kinematics, kinetics and muscle activation during step descent task. Gait posture, 2018. 65: p. 106-111.
38. Redmond AC, Crane YZ, Menz HB., Normative values for the foot posture index. J Foot Ankle Res, 2008. 1(1): p. 6.
39. Martínez-Nova A, Gómez-Blázquez E, Escamilla-Martínez E, Pérez-Soriano P, Gijon-Nogueron G, Fernández-Seguín LM., The foot posture index in men practicing three

ACCEPTED MANUSCRIPT

sports different in their biomechanical gestures. J Am Podiatr Med Assoc, 2014. 104(2): p. 154-158.
40. Cherati AS, Dousti M, Younespour S., Association between foot posture index and ankle sprain in indoor football players. Glob J Health Sci, 2016. 8(10).
41. Willems T, Witvrouw E, Delbaere K, De Cock A, De Clercq D., Relationship between gait biomechanics and inversion sprains: a prospective study of risk factors. Gait posture, 2005. 21(4): p. 379-387.
42. Boling MC, Padua DA, Marshall SW, Guskiewicz K, Pyne S, Beutler A., A prospective investigation of biomechanical risk factors for patellofemoral pain syndrome: the Joint Undertaking to Monitor and Prevent ACL Injury (JUMP-ACL) cohort. Am J Sports Med, 2009. 37(11): p. 2108-2116.
43. Simkin A, Leichter I, Giladi M, Stein M, Milgrom C., Combined effect of foot arch structure and an orthotic device on stress fractures. Foot ankle, 1989. 10(1): p. 25-29.
44. O'Connor, K.M. and J. Hamill, The role of selected extrinsic foot muscles during running. Clinical biomechanics, 2004. 19(1): p. 71-77.
45. Murley GS, Landorf KB, Menz HB, Bird AR., Effect of foot posture, foot orthoses and footwear on lower limb muscle activity during walking and running: a systematic review. Gait posture, 2009. 29(2): p. 172-187.
46. Murley GS, Menz HB, Landorf KB., Foot posture influences the electromyographic activity of selected lower limb muscles during gait. J Foot Ankle Res, 2009. 2(1): p. 1-9.
47. Yazdani F, Razeghi M, Karimi MT, Bani MS, Bahreinizad H., Foot hyperpronation alters lumbopelvic muscle function during the stance phase of gait. Gait posture, 2019. 74: p. 102-107.
48. Khodaveisi H, Sadeghi H, Memar R, Anbarian M., Comparison of selected muscular activity of trunk and lower extremities in young women's walking on supinated, pronated and normal foot. Apunts. Medicina de l'Esport, 2016. 51(189): p. 13-19.
49. Lack S, Barton C, Malliaras P, Twycross-Lewis R, Woledge R, Morrissey D., The effect of anti-pronation foot orthoses on hip and knee kinematics and muscle activity during a functional step-up task in healthy individuals: A laboratory study. Clinical biomechanics, 2014. 29(2): p. 177-182.
50. Mei Q, Gu Y, Xiang L, Baker JS, Fernandez J., Foot pronation contributes to altered lower extremity loading after long distance running. Front Physiol, 2019. 10: p. 573.
51. Jafarnezhadgero A, Alavi-Mehr SM, Granacher U., Effects of anti-pronation shoes on lower limb kinematics and kinetics in female runners with pronated feet: The role of physical fatigue. PloS one, 2019. 14(5): p. e0216818.
52. Resende RA, Deluzio KJ, Kirkwood RN, Hassan EA, Fonseca ST., Increased unilateral foot pronation affects lower limbs and pelvic biomechanics during walking. Gait posture, 2015. 41(2): p. 395-401.
53. Coplan J.A., Rotational motion of the knee: a comparison of normal and pronating subjects. J Orthop Sports Phys Ther, 1989. 10(9): p. 366-369.
54. Bassett C., Lower limb injury and foot posture in international and national level female footballers. 2012.

ACCEPTED MANUSCRIPT

Table 1 - The prevalence of each lower extremity injury among all players, and with a closer look, among all injuries recorded.

	Number of Injuries	Percent of injury among all injuries	Percent of injury among all players
Hamstring	41	26%	17%
Ankle Lig.	34	22%	14%
Ankle Other	2	1%	0.8%
Groin	25	16%	10%
Quad	9	6%	4%
Med.Menisc	4	2%	1%
Lat.Menisc	3	1%	0.8%
ACL	5	3%	2%
PCL	12	8%	5%
MCL	12	8%	5%
LCL	6	4%	2%
Knee Other	2	1%	0.8%

ACCEPTED MANUSCRIPT

Table 2 – The relative risks (95% CI) of three types of lower extremity injuries among pronated, supinated and both together as abnormal foot posture types in professional football players. *: The RRs higher or lesser than one demonstrates the possible effect of the pronated foot on lower extremity injuries

Examination		Hamstring Injuries	MCL Injuries	Absence of fewer than 28 days
		Relative risk (95%CI)	Relative risk (95%CI)	Relative risk (95%CI)
Static Examination	Pronated	**2.1 (1.2-3.6) ***	2.8 (0.9-8.4)	1.3 (1.0-1.7)
	Supinated	1.0 (0.5-2.1)	1.4 (0.4-5.0)	0.8 (0.5-1.2)
	Abnormal	**2.2 (1.2-4.0) ***	4.5 (1.0-19.9)	1.1 (0.9-1.5)
Optic Examination	Pronated	**1.8 (1.1-3.1) ***	2.3 (0.7-6.8)	**1.3 (1.1-1.8) ***
	Supinated	0.9 (0.4-2.0)	1.8 (0.5-6.2)	0.9 (0.6-1.3)
	Abnormal	1.6 (0.9-2.9)	3.5 (1.0-12.5)	1.2 (0.9-1.6)
Foot Posture Index	Pronated	**1.8 (1.1-3.2) ***	2.1 (0.7-6.4)	**1.4 (1.1-1.8) ***
	Supinated	0.8 (0.4-1.8)	1.6 (0.5-5.7)	0.9 (0.6-1.3)
	Abnormal	1.6 (0.9-2.9)	3.1 (0.9-11.2)	1.3 (1.0-1.7)
Visual examination	Pronated	**2.3 (1.3-4.1) ***	4.0 (0.3-14.9)	**1.5 (1.1-2.0) ***
	Supinated	1.3 (0.5-3.2)	2.4 (0.6-10.0)	0.8 (0.4-1.5)
	Abnormal	**2.1 (1.2-3.7) ***	**4.1 (1.6-14.2) ***	1.2 (0.9-1.7)

ACCEPTED MANUSCRIPT

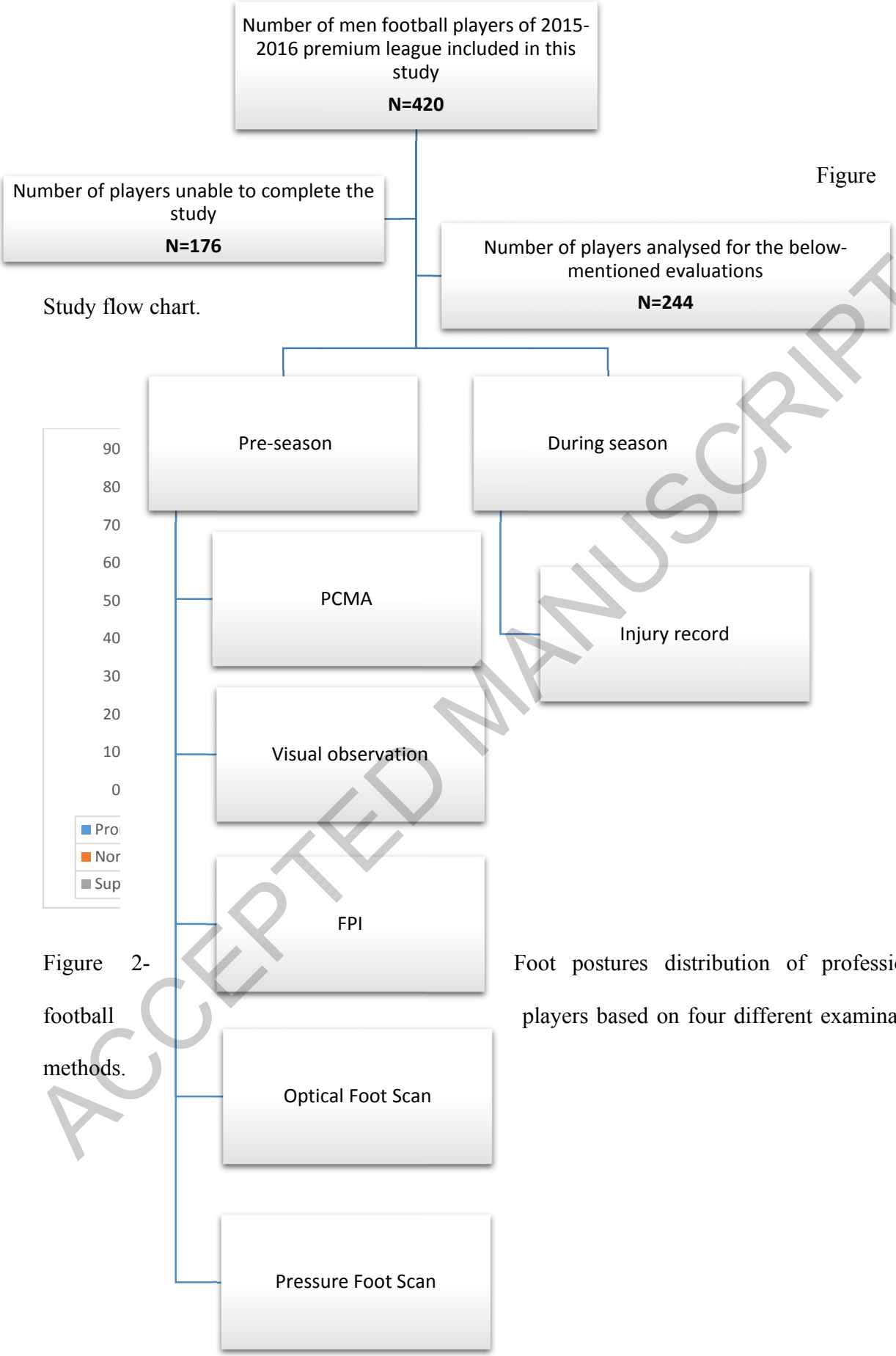


Figure 1- Study flow chart.

Figure 2- Foot postures distribution of professional football players based on four different examination methods.

ACCEPTED MANUSCRIPT

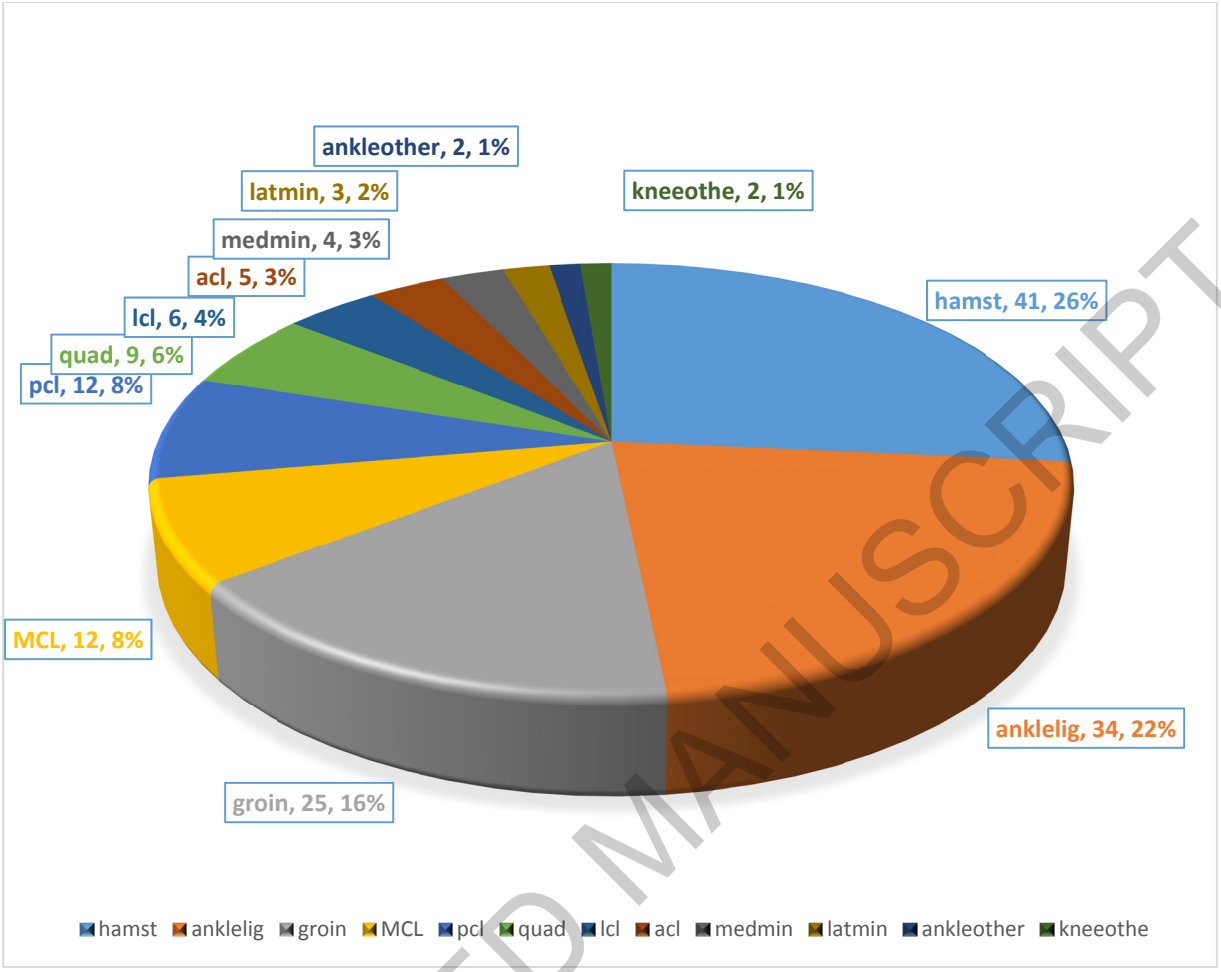
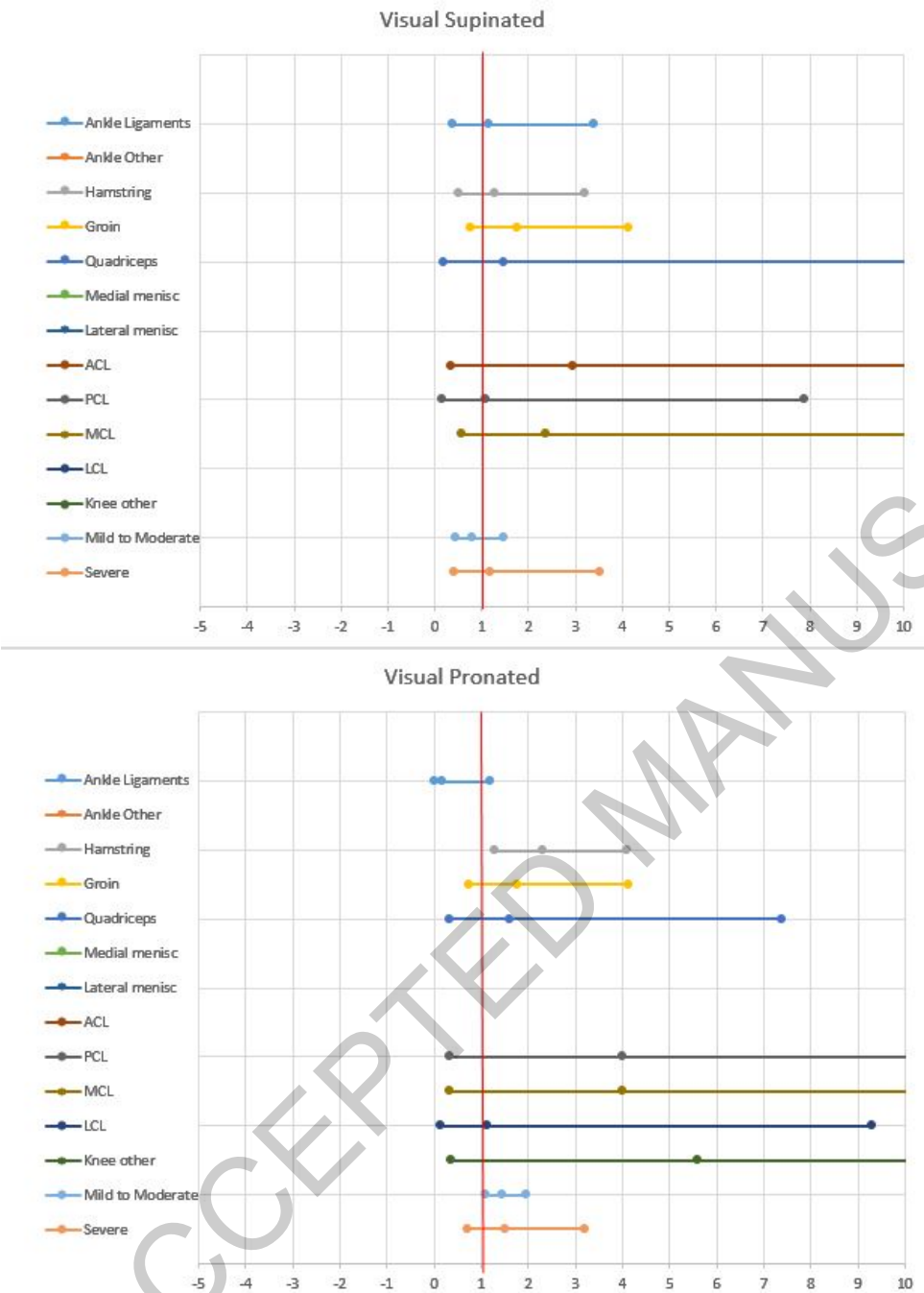


Figure 3 - Injury prevalence among all injuries.

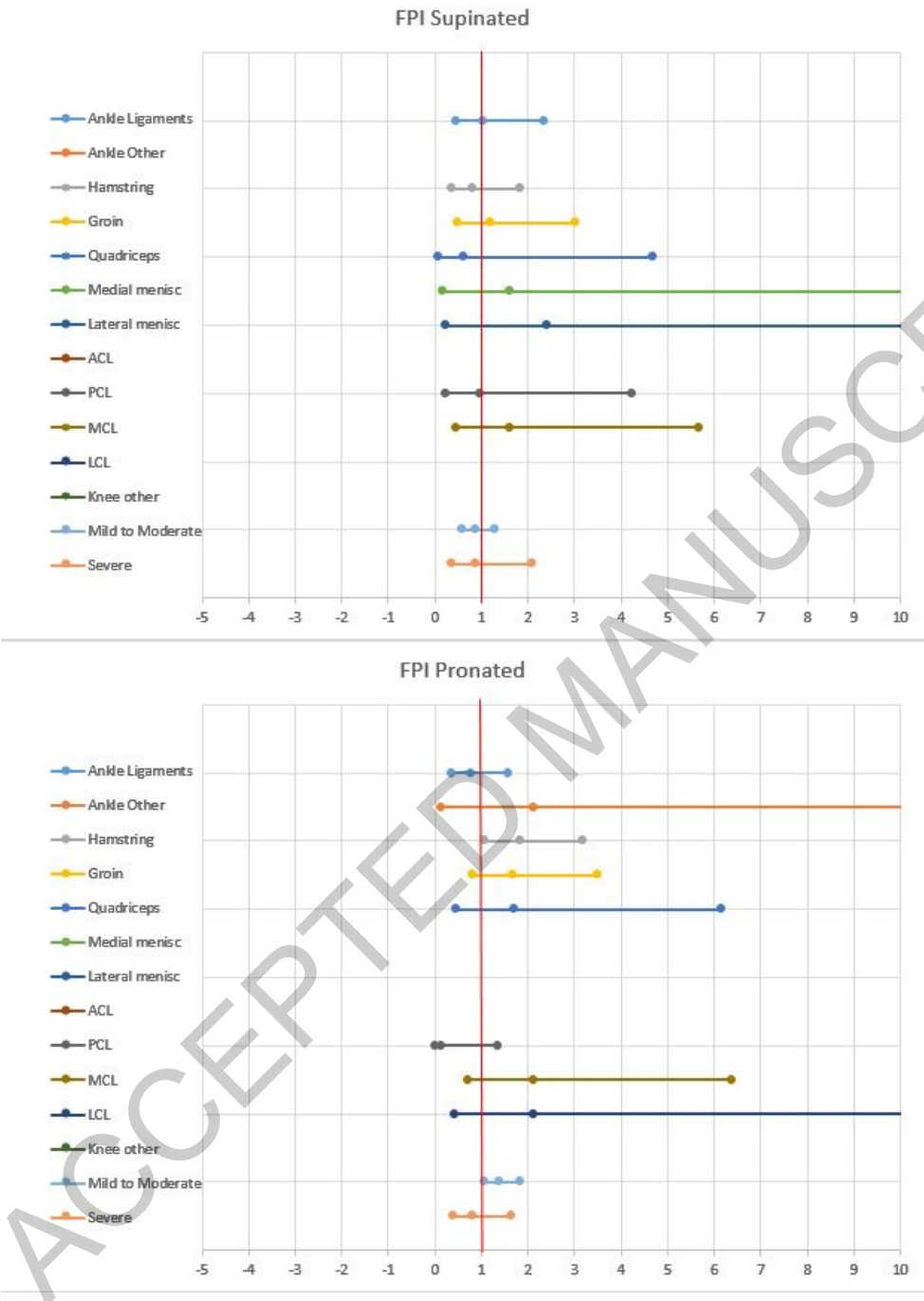
ACCEPTED MANUSCRIPT



ACCEPTED MANUSCRIPT

Figure 4- Relative risk between foot posture based on visual evaluation and lower limb injuries. The dots represent the relative risks of the pronated/supinated foot for mentioned injuries with the confidence interval of 95%. The RR confidence interval of each injury not including one

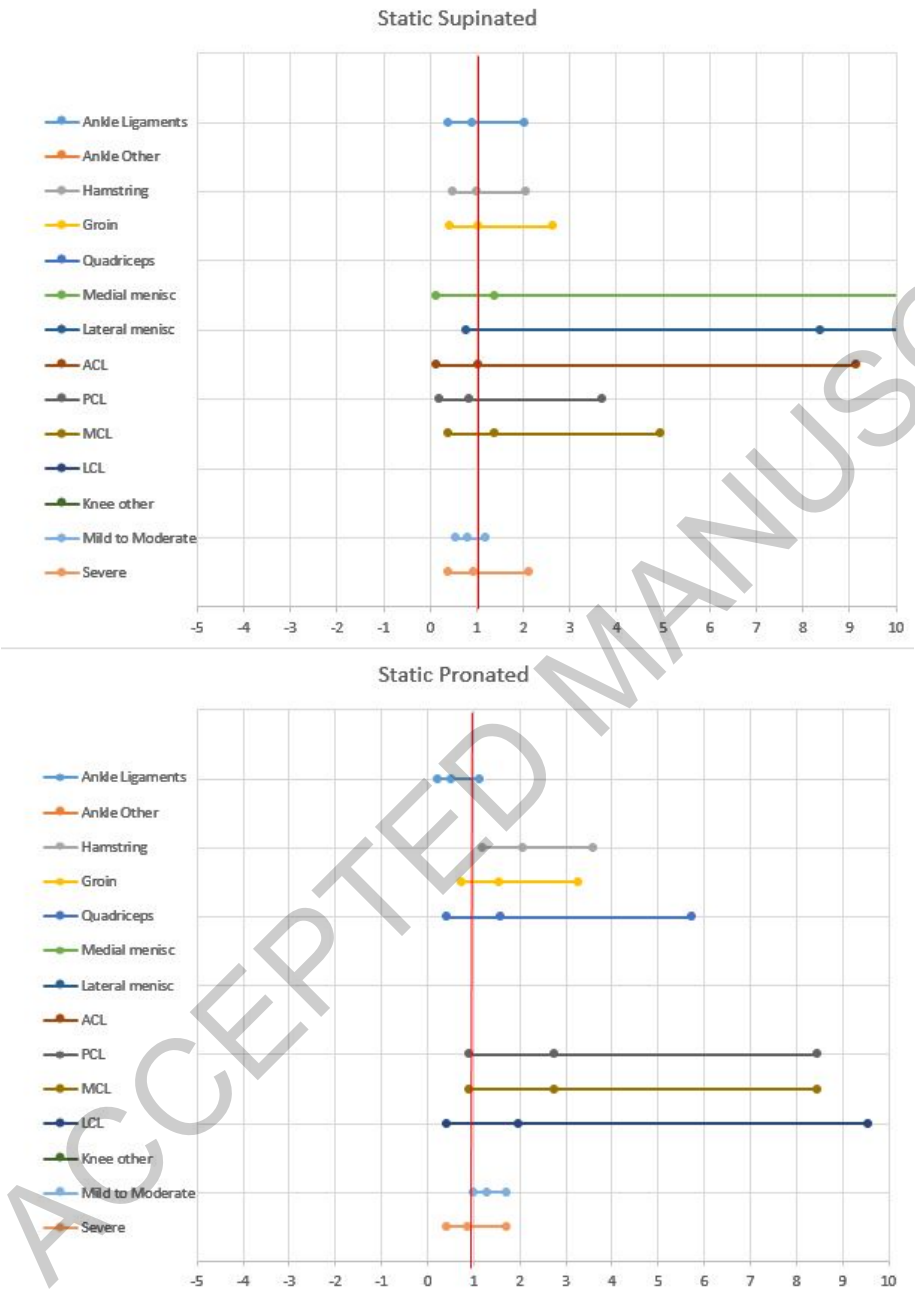
means a statistically significant relationship between the foot posture and mentioned injury.



ACCEPTED MANUSCRIPT

Figure 5- Relative risk between foot posture based on FPI and lower limb injuries. The dots represent the relative risks of the pronated/supinated foot for mentioned injuries with the

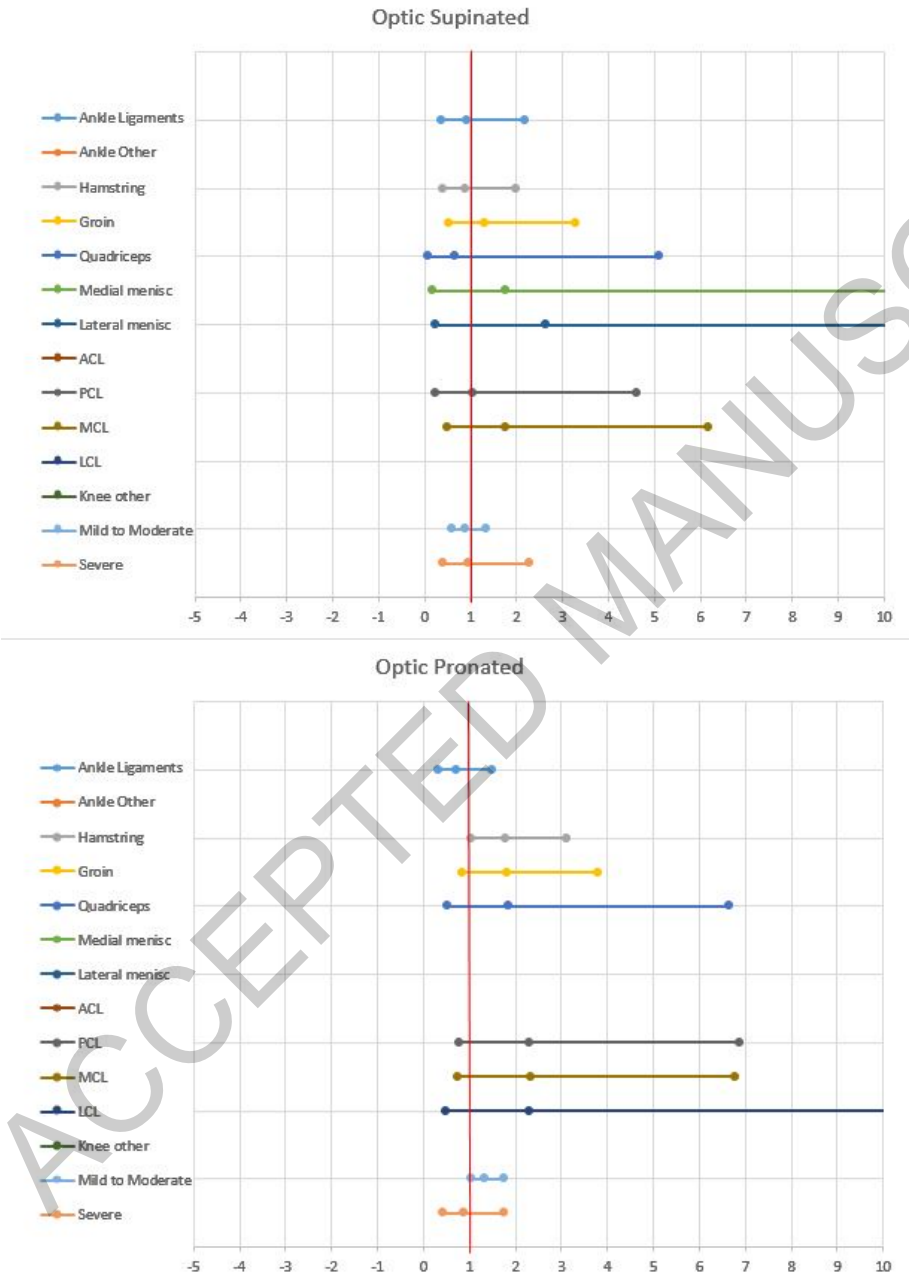
confidence interval of 95%. The RR confidence interval of each injury, not including one, means a statistically significant relationship between the foot posture and that injury.



ACCEPTED MANUSCRIPT

Figure 6- Relative risk between pronated foot posture based on static Device evaluation and lower limb injuries. The dots represent the relative risks of the pronated/supinated foot for

mentioned injuries with the confidence interval of 95%. The RR confidence interval of each injury, not including one, means a statistically significant relationship between the foot posture and that injury.



ACCEPTED MANUSCRIPT

Figure 7- Relative risk between pronated foot posture based on optic Device evaluation and lower limb injuries. The dots represent the relative risks of the pronated/supinated foot for mentioned injuries with the confidence interval of 95%. The RR confidence interval of each injury, not including one, means a statistically significant relationship between the foot posture and that injury.

ACCEPTED MANUSCRIPT